

PathPilot AI

AI-Powered Personalized Learning Path Recommender — Technical & Submission Document

1. Solution Overview

PathPilot turns a learner's natural-language career goal, current skills, interests, experience, learning preference and study capacity into a structured learning roadmap. It identifies skill gaps, respects prerequisites, ranks resources, explains recommendations and adapts using progress, assessments and feedback.

2. Architecture

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Browser UI → FastAPI → NLP/Profile Engine → Skill Gap + Proficiency Engine → Semantic Retrieval → Hybrid Ranking → Prerequisite-aware Path Generator → Progress/Feedback Loop
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3. Recommendation Intelligence

The recommender is deliberately hybrid rather than a simple chatbot. TF-IDF n-gram vectors and cosine similarity provide semantic matching. An interpretable scoring model combines competency priority, semantic relevance, prerequisite readiness, experience fit, interest match, resource quality, learning preference and feedback. One primary resource is selected per competency to prevent duplicates from crowding the roadmap.

4. Personalization & Adaptation

Explicit proficiency words such as basic, beginner, advanced and expert are detected. Mastered skills are filtered out. Assessment scores update mastery; scores below 70% keep the learner in reinforcement mode. Completion removes resources from future recommendations. Helpful, save, skip and not-helpful feedback changes ranking.

5. Conversational Assistant

The assistant accepts natural-language goals and questions. The local engine can answer goal, skill-gap, recommendation, timeline and next-step questions without an API key. An optional LLM layer can enrich responses when configured, while structured recommendations remain the source of truth.

6. Evaluation

The repository includes an offline benchmark with 18 representative learner scenarios and reports Precision@3, Recall@3 and NDCG@3. On this curated validation set, the current engine should be described using the measured benchmark output rather than as a universal real-world accuracy figure. A production evaluation should use held-out learner interactions and expert relevance labels.

Component	Implementation
Frontend	Responsive HTML/CSS/JavaScript dashboard
Backend	Python + FastAPI
Recommendation	TF-IDF + cosine similarity + hybrid weighted ranking
Adaptation	SQLite progress, assessment and feedback signals
AI	Optional LLM response layer + deterministic structured recommender

7. Demo Scenario

Input: "I want to become a backend developer. I know Java and basic SQL." PathPilot detects the goal and existing skills, skips mastered competencies, identifies REST APIs → Spring Boot → JPA/Hibernate → Docker → System Design as remaining gaps, explains each recommendation and estimates completion time from weekly study capacity.

8. Validation Commands

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PYTHONPATH=. python evaluate.py | PYTHONPATH=. pytest -q
```

9. Submission Notes

The prototype is designed for a hackathon/academic submission. The benchmark score is an internal validation result and should not be presented as a claim that the system is 100% accurate for arbitrary real learners. The strongest demo is the adaptive loop: profile → recommendation → explanation → assessment/completion → changed roadmap.